Electrical Energy

Electricity is a form of energy where the energy supply comes from rapidly moving electrons. You may remember from year 8 that there are two main forms of electrical energy;

- Static electricity
- Current electricity

Try this exercise to reacquaint you with static electricity

A shocking experience



Imagine you and three friends are camping in the country in the middle of a cleared paddock. You have driven here the evening before. Your car is parked outside a gate into the paddock. After lunch, you fall asleep in the warm sunshine. You awaken several hours later to the sound of a loud crash. The sun is hidden by a huge thunder cloud. You see a flash of lightning and 12 seconds later, you hear a loud crack of thunder.

Your gear is spread out around you on a blanket. It includes a picnic basket, plastic plates, stainless steel knives and forks, a metal Esky with plastic cover, remains of food and drink, a garbage bag and beach umbrella. You have driven the metal pole of the beach umbrella deep into the ground so that it shades the food. Your tent is nearby.

You are dressed in shorts and T-shirts. Each of you has a watch. Your watch is plastic but your three friends have metallic watches. One girl has gold metal clips in her hair and one of your friends is wearing long dangling silver earrings. There are some trees in the distance and a small metal shed hidden amongst the trees. There is a single large gum tree nearby.

The storm is approaching rapidly from the east, the thunder is very loud and the lightning is very bright.

An exercise in survival

How much time have you got before the storm is overhead?
What are you going to do?

1. **In a group, decide upon the most sensible plan of action. Write down your plan, explaining the reasons behind your actions.**

2. **Did you have all the information you needed to make a sensible decision?**
   What more would you have liked to know?
   Often we have to make decisions on incomplete evidence. The more background information you have, however, the more likely your decisions will be the best possible in the circumstances.

3. **Before lunch, there was no indication that your lives were threatened. You were safe and healthy and had a 'survival rating' of 100.**
   What would your group's survival rating be after the storm?
   Look up the sheet 'Survival ratings'.

4. **Do you agree with the ratings given on the sheet or would you swap some actions around?**
   Explain.

1. STATIC ELECTRICITY:

SOURCES OF ELECTRICAL ENERGY

A convenient source of electrical energy in the laboratory is the cell. The electrons the cell releases come from chemical reactions that occur within it. That is, cells are electron sources, and they can 'pump' electrons around an electrical circuit.

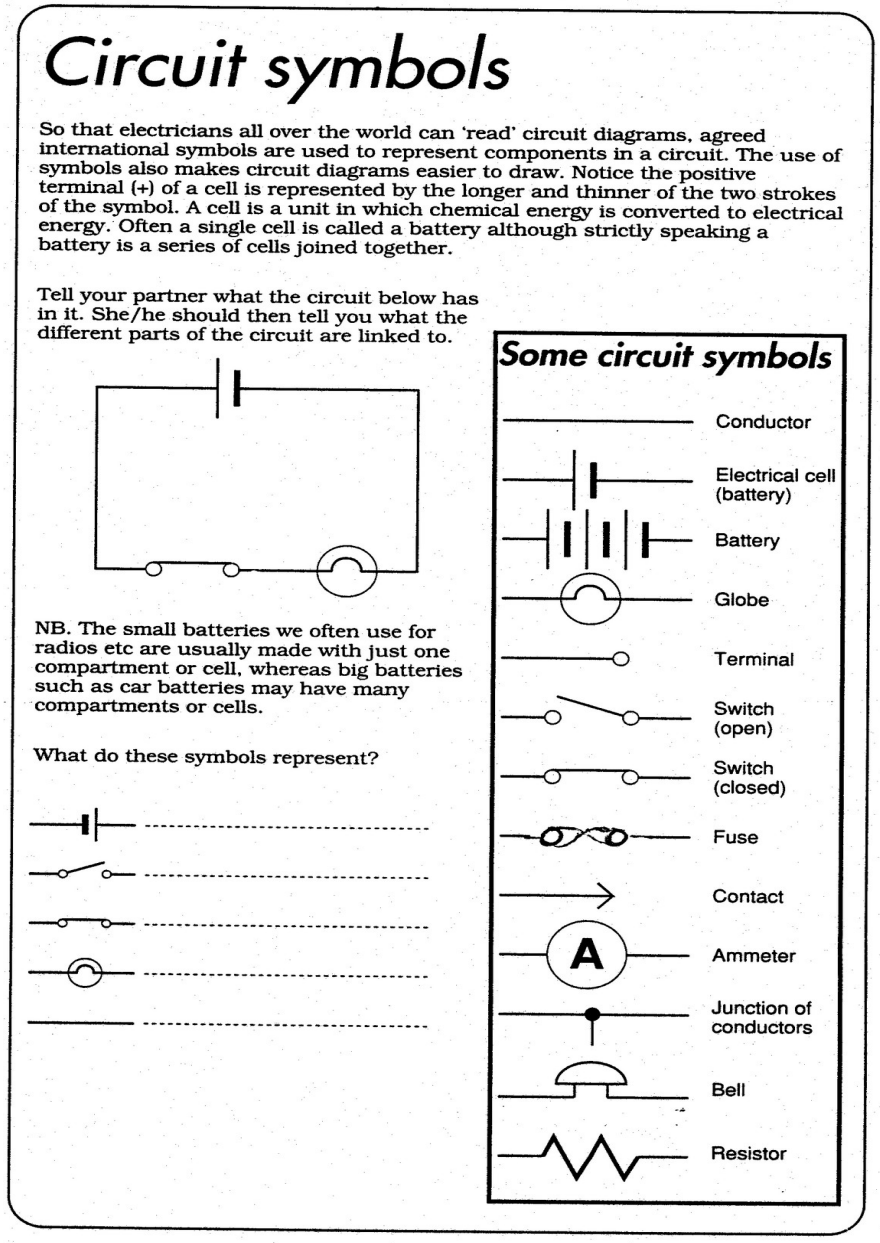
A battery is made up of two or more cells linked together.

2. CIRCUITS

a) Simple Circuits

A circuit is a completed loop of conducting material joining the terminals of a cell or a battery.

In the drawing of circuits, the following symbols are used:



A circuit is drawn using straight ruled lines and in a rectangular shape even though in reality it may look quite different.

Draw examples of circuits in the space below.

i) a cell, a switch, a globe ,an ammeter, a resistor joined by conducting wires.

ii) 2 cells, 2 globes, a bell and a resistor joined by conducting wires

b) Electric Current

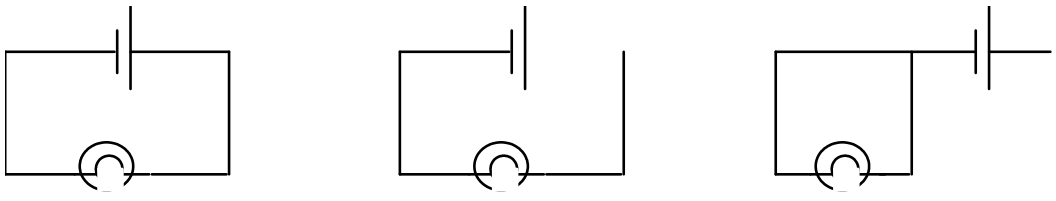
An electric current is a flow of charged particles, pumped by a cell, around a circuit. In metals, the electric current is a flow of electrons. For a current to flow, the circuit must be complete.

The size of current flowing around a circuit can be determined by examining the brightness of a globe in the circuit (the brighter the globe, the greater the current).

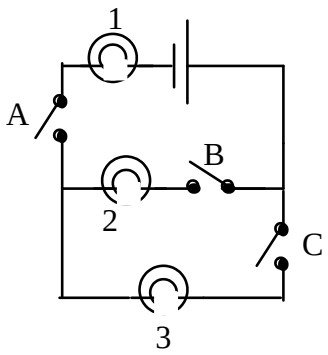
A switch is a device for breaking a circuit. I can be drawn open or closed.

Questions

1. In which of the following will an electric current flow?



2. Consider the following circuit:



Which globes will glow when the following switches are on?

i) Switch A
ii) Switch A and Switch C
iii) Switch B and Switch C
iv) Switch A and Switch B

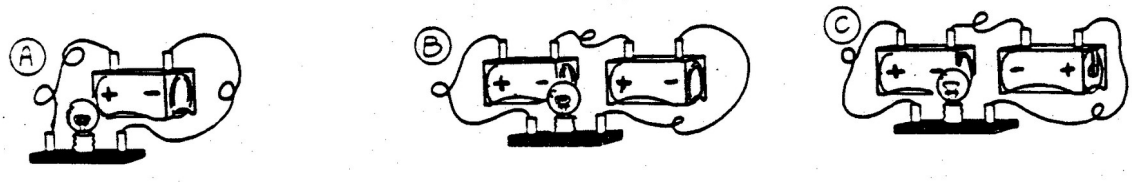
3. Now you draw a circuit in which there are 3 globes and 2 switches and 2 cells. When 1 switch is closed only 1 globe glows and when the other switch is closed the 2 other globes glow.

ACTIVITY 1: Simple Circuits

Materials: 1 electricity kit
Examine all the components of the kit carefully and make sure you understand what each is.

Method:
1. Build each of the following circuits and determine whether the globe is off, dim or bright.

Draw each of the circuits and write your answer under each circuit diagram..

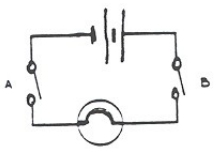


2. Using 2 cells, 1 switch, 2 globes and conducting wires, build as many different circuits as you can, in which the light globes glows when the switch is on.

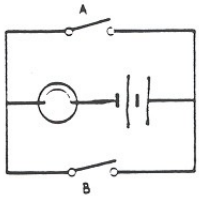
Draw each of these circuits in the space below..

3. Complete the tables by either connecting up each circuit, or by "tracing" the electron flow around the circuit.

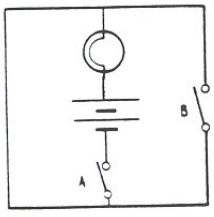
Put a √ for lamp ON
and a X for lamp OFF



Switches		Lamp
A	B	
OFF	OFF	
OFF	ON	
ON	OFF	
ON	ON	



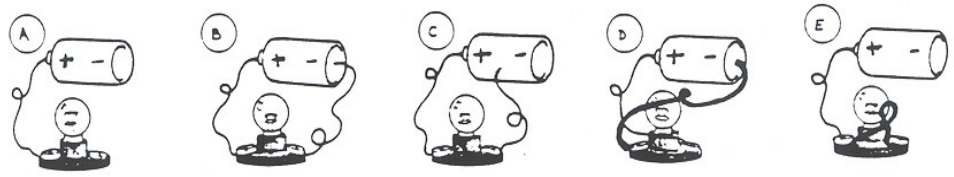
Switches		Lamp
A	B	
OFF	OFF	
OFF	ON	
ON	OFF	
ON	ON	



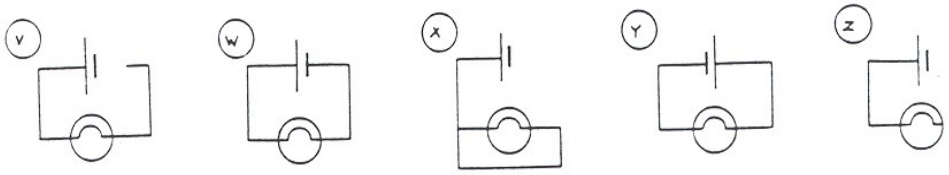
Switches		Lamp
A	B	
OFF	OFF	
OFF	ON	
ON	OFF	
ON	ON	

PROBLEM SHEET 1 – Simple Circuits

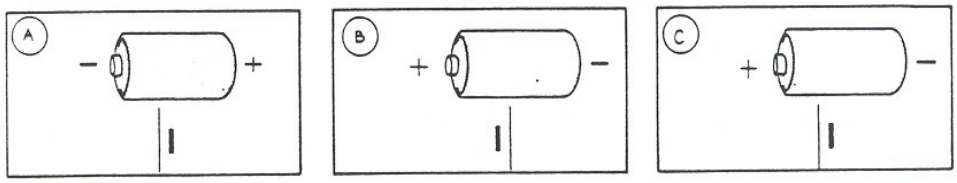
1. Which globes would light?



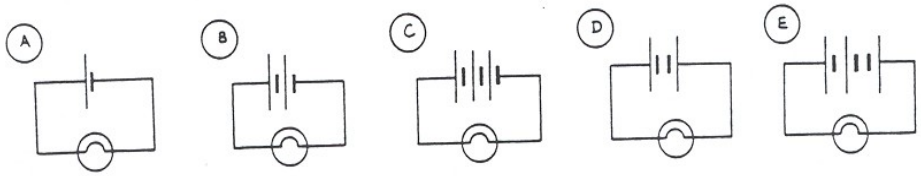
2. Below are circuit diagrams for the circuits in question 1., but in different order. Which diagram goes with each globe circuit?



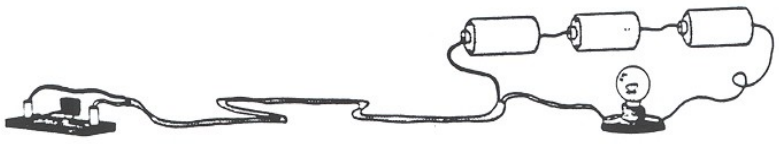
3. Which box contains a cell with its terminals marked correctly and its circuit symbol drawn the same way round as the cell?



4. Which globe would be the brightest?



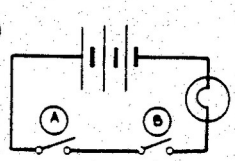
5. Draw the circuit diagram for this circuit:



6. Complete the tables for the switches in the following circuits:

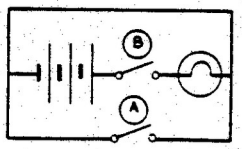
✗ = lamp OFF
✓ = lamp ON

(i)



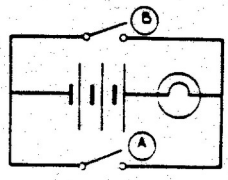
Switches		Lamp
A	B	
OFF	OFF	
OFF	ON	
ON	OFF	
ON	ON	

(ii)



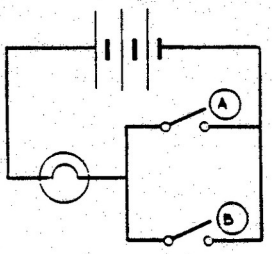
Switches		Lamp
A	B	
OFF	OFF	
OFF	ON	
ON	OFF	
ON	ON	

(iii)



Switches		Lamp
A	B	
OFF	OFF	
OFF	ON	
ON	OFF	
ON	ON	

7. This circuit is the same (electrically) as one of the circuits in question 6. Find out which it is by working out the switch table for this circuit and comparing it with the ones in question 6.



Switches		Lamp
A	B	
OFF	OFF	
OFF	ON	
ON	OFF	
ON	ON	

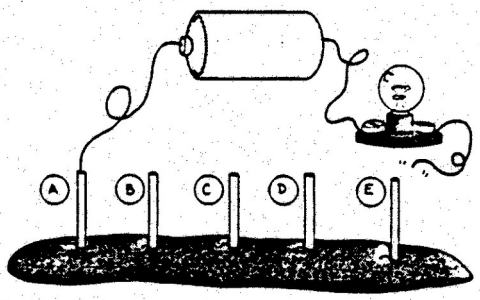
FOR THE EXPERTS:

8. A student knew that two U-shaped pieces of copper and one 'odd end' had been buried in plasticine. She used a lamp circuit to test each end and noted her results as follows: AB meant her test circuit was connected to A and B; -AB- meant that this connection lit the lamp.

Here are the results:

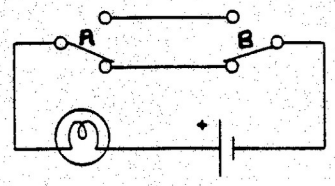
AB
AC BC
-AD- BD CD
AE -BE- CE DE

Which is the odd end?
Which ends are connected together?



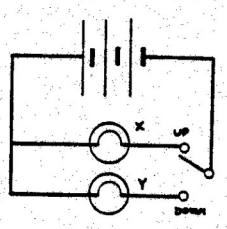
9. a) The lamp in the circuit shown below can be put on by one switch and off by another switch. (The switches used are called two-way switches; each can connect first to one wire then another).

Put 'on' or 'off' for the lamp, for each situation described.

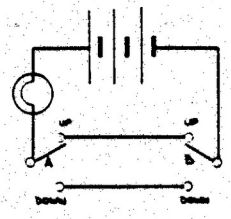


Switch A	Switch B	Lamp
up	down	
down	down	
down	up	
up	up	

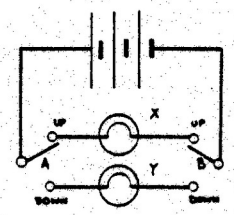
b) Three other circuits using two-way switches are given below. Complete the switch table for each circuit.



Switch	Lamps	
	X	Y
UP		
DOWN		



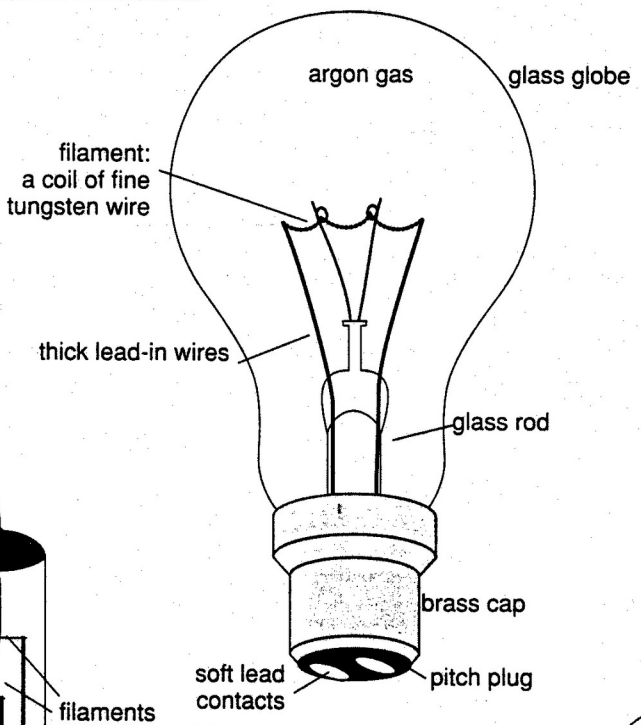
Switches		Lamp
A	B	
UP	UP	
UP	DOWN	
DOWN	UP	
DOWN	DOWN	



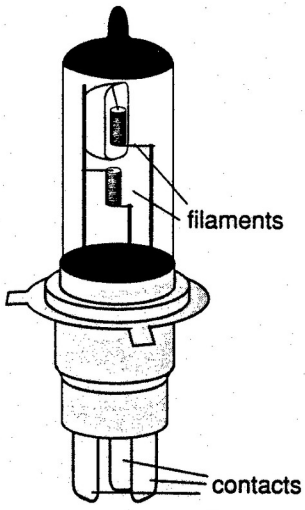
Switches		Lamps	
A	B	X	Y
UP	UP		
UP	DOWN		
DOWN	UP		
DOWN	DOWN		

Some examples of incandescent lights

1. For each part of the household light globe labelled in the diagram below, write down what its function is.

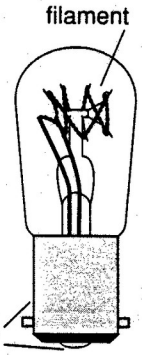


Household light bulb

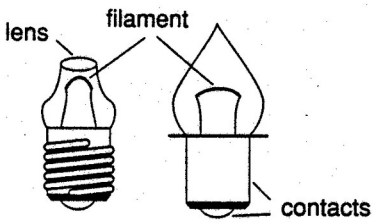


Car headlight bulb

note the two filaments, one for high beam and one for low beam



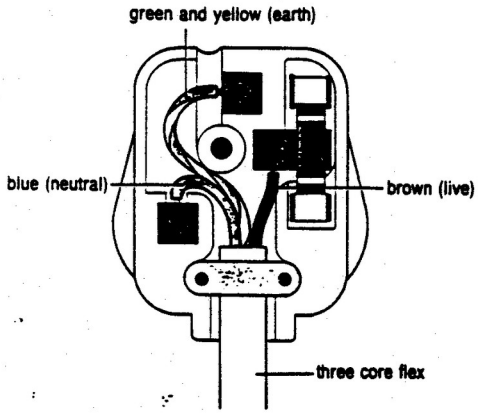
Refrigerator light

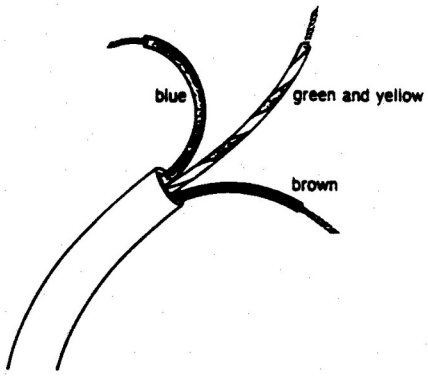


Torch globes

Electricity in the home

Read the following passage and answer the questions which follow.





Three core flex connected to a plug. The cores are made from many strands of thin copper wire to permit movement and bending, hence the name of flexible cord or 'flex'. The insulation may be PVC, rubber or braided textile.

Flexes and fuses

Many sizes and types of flexible cord are used to connect household appliances to the mains. Appliances with bigger power ratings need thicker flexes and bigger fuses in the plugs. You can see why if you take hold of a flex which is supplying electrical energy to a 3 kW electric fire. The flex may feel warm. In fact all wires become slightly warm when there is a current in them, but the heating is greater with large currents.

Plug fuses provide some safety. If a fault develops which causes the current to become too large the fuse will melt and break the circuit. The current stops, the flex no longer overheats and the risk of fire is reduced.

For every appliance you use it is important therefore to:

- fit a thick enough flex to carry the current without overheating;
- choose the lowest value of fuse (usually 3A or 13A) which will carry the normal current but will melt if the current becomes accidently large.

Power rating (W)	Flex thickness (mm^2)	Plug fuse (A)	Typical appliance
up to 720W	0.5	3	table lamps food mixers clocks
720 to 1440W	0.75	13	vacuum cleaners colour televisions irons toasters
1440 to 240W	1.0	13	kettles fan heaters
2400 to 3240W	1.25	13	3kW fires



Questions

1 Why do the wires in an electrical cord need to be thicker when connected to an electric room heater than when connected to a table lamp ?

2 How does the plug fuse reduce the risk of fire ?

3 What is the most common cause of a fuse "blowing"?

4 Why is the **lowest** value fuse wire chosen to protect a circuit carrying a normal current ?

5 Some electrical cords on appliances have two wires instead of three. Which wire would be missing on the "two wire" cord ?

6 In what circumstances is it safe to use a "two wire" cord instead of a "three wire" cord ?

7 What is the function of the earth wire ?

8 Why is the external casing of an electrical cord (flex) made of PVC, rubber or textile ?

9 A three wire cord has three wires capable of carrying an electric current. Name the three wires and describe the function of each wire.

10 From the table show, what plug fuse would you fit to appliances of power rating:

(a) 400 W (b) 2 kW (c) 100 W ?

11 From the table show, what thickness of flex would you connect to:

(a) an iron (b) a vacuum cleaner (c) a toaster

Are you safe at home?

Take this sheet home and work through it with your family.

Answer each question with Y for Yes or N for No

1. Do you and your family:

- leave switches on when removing power plugs from sockets? ___
- use any available fuse wire in fuses? ___
- pull the cord, not the plug, to remove a plug from a socket? ___
- take portable electric radiators or heaters into the bathroom? ___
- use frayed or damaged cords? ___
- use more than one double adaptor in a power point? ___
- use a cord with twin flex (two wires together) as an extension cord? ___
- fly kites and model planes near overhead wires? ___
- use a plug socket adaptor (piggy-back plug) as a cord extension socket on extension cords? ___
- use detergents to clean switches and sockets? ___
- touch electrical appliances or cords with wet hands? ___
- keep your hair and cosmetic appliances in the bathroom plugged in when you are not using them? ___
- forget to protect small children from putting things in power points by having the power point high up or using protective caps? ___
- use old unapproved plugs? ___
- allow tree branches to grow close to electrical wires? ___

2. In your household, do you or other members of your family know the location of:

- the connection to your home from the mains electricity supply? ___
- the connection to your home for the telephone line? ___
- the main earth wire connected to a water pipe or earth stake? ___
- underground electrical cables in your garden? ___
- the fuses or circuit breakers for your home? ___
- the switch to turn off power to the power points, the stove, and the lights? ___

3. Draw a plan of your house and garden showing the location of the items mentioned in the question above.

4. An electric current of 0.07 amperes is enough to kill you. Examine an electric appliance such as a food blender at home. Make sure you pull the plug out of the power point first. If you look at the back you may find a small plate with details about the electrical specifications of the appliance. See if you can find any information about how much electrical current it uses. Look for numbers with A or amps for amperes after them.

Type of appliance studied

_____

Current used _____

3. RESISTANCE

Resistance in a circuit is any device that restricts or inhibits the current through it.

1. If a device e.g. a piece of wire, has a high resistance then the flow of electricity through it is greatly reduced.

2. If a device has a low resistance then the flow of electricity through it will be only slightly reduced.

3. The greater the resistance of a device, the smaller the current that will flow through the circuit.

4. When an electric current flows through resistance in a circuit, some of the electrical energy is transformed to heat energy i.e. heat is produced when electricity flows through resistance.

5. The greater the resistance, the greater the amount of heat produced.

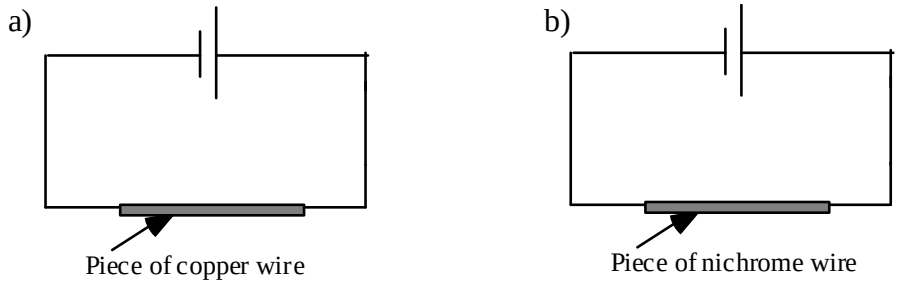
6. Different conductors can have different resistance properties.

   For example, copper has a low resistance but nichrome and tungsten have high resistance.

Light globes contain a piece of high resistance wire, which produces much heat (and thus light) when an electric current is passed through it. Thus, a light globe can be regarded as a device with high resistance.

Example

1. Which of the following circuits has the greater resistance?



(Both pieces of wire are of the same length and diameter)

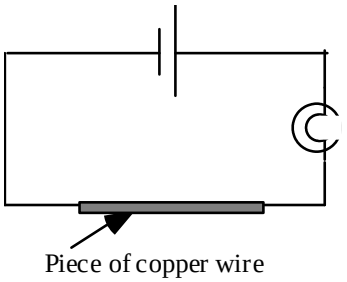
2. Which of the following circuits has the lower resistance?



ACTIVITY 2: Resistance

Method:

1. Set up the following circuit



1. How bright is the globe ? _____

2. Replace the copper wire with a piece of nichrome wire (of the same length as the copper wire). How bright is the globe now ? _____

3. Compare the brightness of the globe with that obtained for the copper wire.

_____

4. Which wire has the greater resistance, copper wire or nichrome wire? _____

5. Now replace the nichrome wire with a small piece of 'steel wool' wire. Describe what happens to the globe now.

_____

6. Explain what happened with the 'steel wool wire' . _____

_____

7. Electrical energy is more easily transformed to heat energy using _____ resistance wire.

Application of Resistance

Offering resistance

If you are caught outside in a lightning storm with no shelter are you better off with wet or dry clothes? Water is a fairly good conductor of electricity. Surprisingly, the answer is wet clothes. If your clothes are dry they offer more resistance to the flow of electricity than your body which contains quite a lot of water, so the lightning will pass through your body rather than your clothes. If your clothes are really wet then your body will offer more resistance to electricity than your clothes. The lightning will pass through your clothes rather than you and this might save you from severe burns or even death. This is not, however, to be relied upon when in danger of being struck by lightning!

If electricity has a choice of two pathways it will always travel along the one which offers it least resistance. Electrical resistance (R) is a measure of how difficult it is for an electric current to pass through a substance. Just as mass is measured in kilograms and height in metres, resistance is measured in ohms. The symbol for the Greek letter omega (Ω), is used as the symbol for ohms. Electrical resistance is measured with an ohmmeter or multimeter.

Some examples of resistance

Range of normal skin	10 000-3 000 ohms
Wet skin	300 ohms
Skin soaked in salt water	100 ohms
Light bulb	70 ohms
Electric toaster	15 ohms
Electric frypan	10 ohms
Electric lamp cord	0.3 ohms
Fuse wire	2 ohms

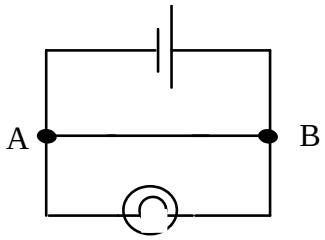
The resistance of your skin can vary quite a lot depending on how moist it is. Electrical resistance is the secret of the 'lie detector'. The suspect's hand is hooked up to the device which records her/his electrical resistance. The suspect is then asked questions such as: 'Where were you at the time of the crime?'

1. What happens to the palms of your hands when you become nervous?
2. If the suspect lies, why might the resistance be lowered?
3. Sometimes, innocent people can show low resistance when questioned. Can you think of a reason for this?

4. SHORT CIRCUITS

A 'short circuit' is an alternative conducting pathway of lower resistance, which by-passes a particular resistance, globe or other appliance in a circuit. It is not necessarily the shortest path however.

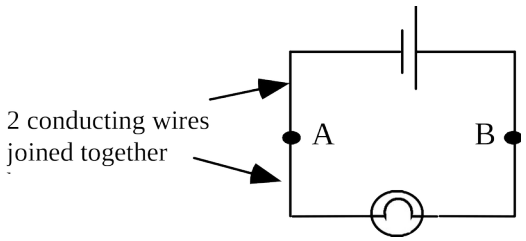
E.g. in the following circuit, the light globe will not light up because of the 'short circuit' between points A and B (i.e. it is 'easier' for the electrons to travel through wire AB rather than through the light globe.)



ACTIVITY 3: Short Circuits

Method:

1. Set up the following circuit



2. Sides A and B are made of two wires each, joined together by twisting the ends

3. Place another conducting wire between A and B. This is the short circuit.

4. Predict what you think will happen. _____

Test your prediction.

5. What happens to the globe when the wire between A and B is connected ?

_____

6. Explain what happened to the globe. _____

_____

5. FUSES

Fuse wire is a relatively high resistance conductor with a low melting point.

A fuse wire will melt if too large a current is passed through it. When it melts the circuit is broken and current can no longer flow.

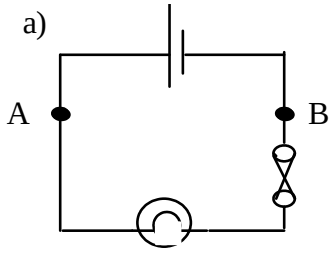
A fuse is a protective or safety device, included in a circuit, which melts when a short-circuit or overload
results in too high a current passing through the circuit.

In circuit diagrams, a fuse has the symbol

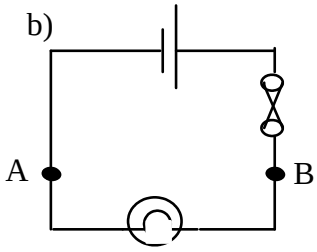
Example

In which of the following circuits would you expect the fuse to blow if a wire was placed between A and B?

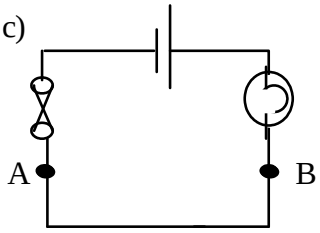
a)



b)



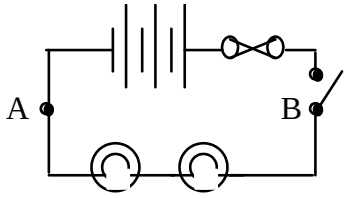
c)



ACTIVITY 4: Fuses

Method:

1. Connect up the following circuit using a strand of steel wool as the fuse wire. Twist the steel wool repeatedly to make the steel wool wire. Press the switch on.



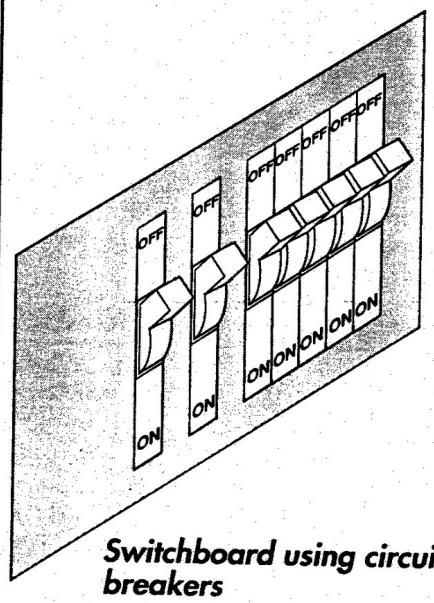
2. Do the globes glow? _____

3. Where would you place a wire to create a short circuit ? Press the switch on, then short circuit the globe circuit.

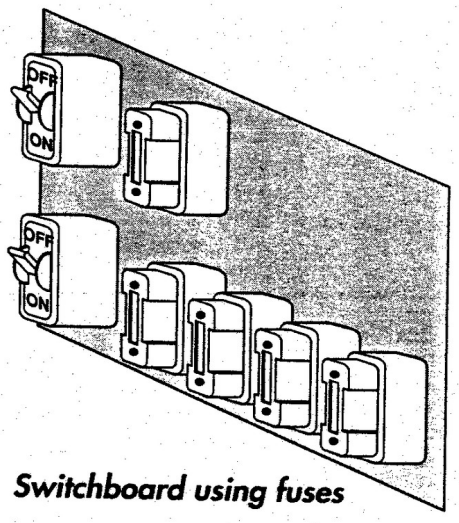
What happens this time ? _____

4. Explain *why* this happens. _____

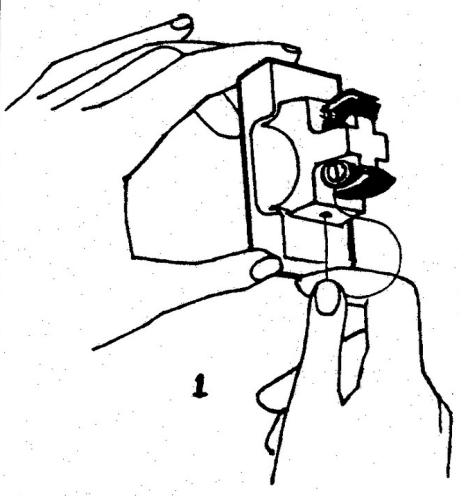
Fuses and circuit breakers

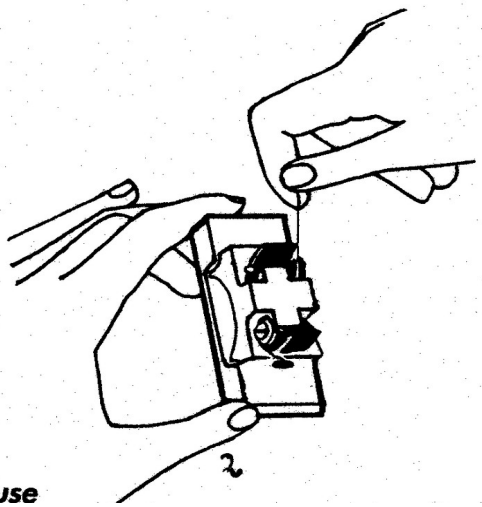


Switchboard using circuit breakers



Switchboard using fuses





Rewiring a fuse

Read this passage and answer the questions on the following page.

Too much of a good thing

Elizabeth and Peter extended their house. They couldn't afford to have a new power circuit installed, so they asked an electrician to connect the power points in the new rooms to the old circuit. Because they had extra rooms they used more electrical appliances. On one busy winter's night, they were operating the following appliances on the one circuit:

- a one-bar electric radiator
- a dishwasher
- a washing machine
- a dimplex oil-filled heater
- a computer
- a clothes dryer.

Elizabeth was typing merrily at the computer when she felt like having a cup of coffee. Suddenly, after the electric jug was switched on, everything stopped. The computer screen went blank. Elizabeth and Peter then had to turn off at least one of the appliances. They had to go outside and flip the circuit breaker switch back to the 'on' position. As long as they turned off one of the appliances they were using beforehand, they could boil the water for their cup of coffee. They have both learned to save their computer files before making coffee as well!!

1. What do you think happened and why?
2. Why was it important for Elizabeth and Peter to press 'save' on the computer before making coffee?
3. How would the situation be different if Peter and Elizabeth had fuses in their meter box instead of circuit breakers?
4. Would there be any problem in replacing a melted fuse wire for a power circuit with fuse wire for a stove circuit that could take up to 30 amps of current before it melted?



1. _____

_____

2. _____

_____

3. _____

_____

4. _____

_____

PROBLEM SHEET 2 - Resistance, Short Circuits and Fuses.

1. Identify the following statements as **true or false**

a) An electric current can flow more easily through a wire of high resistance than through a wire of low resistance.

b) Wire with high resistance restricts the flow of electricity through it.

c) If a wire has a low resistance, it will probably heat up considerably when a current is passed through it.

d) A 'short circuit' is a circuit which is very small.

e) A 'short circuit' can occur if in a particular circuit an alternative pathway of low resistance is produced.

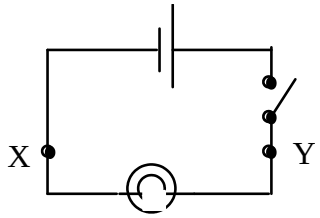
f) Fuse wire is made from wire which has low resistance and low melting point.

g) Fuse wire is made from wire which has high resistance and high melting point.

h) Nichrome wire has a higher resistance than copper wire.

i) The wire used to make the filament in a light globe has high resistance and low melting point.

2. Explain why the globe in the following circuit will go out when a wire is placed between points X and Y.

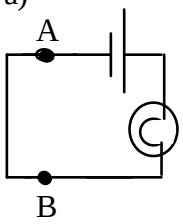


_____

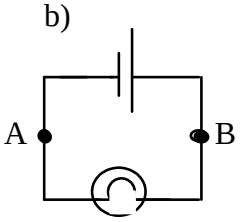
_____

3. Which of the following circuits will be 'short circuited' (i.e. the globe will stop glowing) when a wire is placed between points A and B?

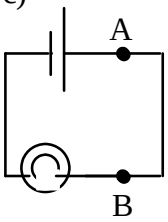
a)



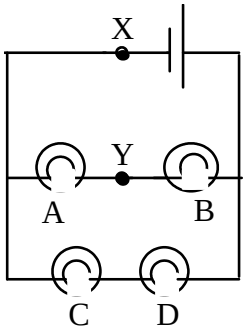
b)



c)



4. For the following circuit, which globe/s will go out when a wire is placed between X and Y?



6. SERIES CIRCUITS

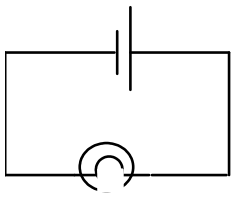
a) A series circuit is one, which contains **only one** possible pathway through which the current can flow.

Example

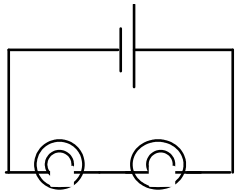
Draw a circuit diagram in the space below in which the two cells are in series and the two globes are in series

b) When two globes are in series in a circuit, if one globe is removed, the other <u>will not</u> light.

c) Two globes in series will not glow as brightly as just one globe i.e. the globe in circuit A will glow brighter than the two globes in circuit B (assuming the other parts of the circuits are the same).



Circuit A



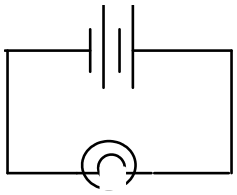
Circuit B

Example
Suggest reasons why the globes in Circuit B will be dimmer:

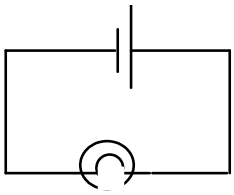
_____

_____

d) Two cells in series will provide twice the electron 'pumping' energy (or voltage) of one cell. For example, the globe in circuit C will glow brighter than the globe in Circuit D.



Circuit C



Circuit D

Example
Suggest reasons why the globe in Circuit C will be brighter:

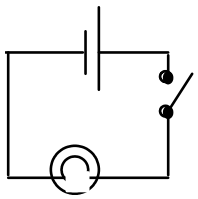
_____

ACTIVITY 5: Series Circuits

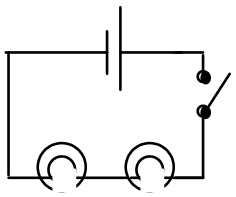
Method:

1. Construct the following circuits and record the relative brightness of the globes on the diagram.

A



B



Brightness ? _____ _____

2. Using the globe brightness as a guide in which circuit was the larger current flowing?

_____

3. Which circuit had the greater resistance? _____

4. Describe the components in the two circuits and the way in which they are connected in words in the space below:

Circuit A contains _____

Connected in _____

Circuit B contains _____

Connected in _____

5. Construct ***circuit B*** again. Push the switch on, and then remove one of the globes from its socket. What happens to the remaining globe?

_____

6. ***Explain*** why this happens to the other globe. _____

_____

7. Construct the following circuits and note the relative brightness of the globes in each

Circuit C - contains 1 cell, a switch and 2 globes in series

Circuit D - contains 2 cells in series, a switch and 2 globes in series

8. Draw a circuit diagram for each of the circuits and underneath each record the relative brightness of the globes

Circuit C ***Circuit D***

7. PARALLEL CIRCUITS

a) A parallel circuit is a circuit with **more than one** pathway through which the current can flow.

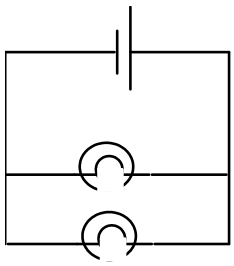
Example
Draw a circuit diagram in the space below in which the two cells are in parallel and the two globes are in series.

b) When two globes are in parallel, if one globe is removed, the other globe continues to glow.

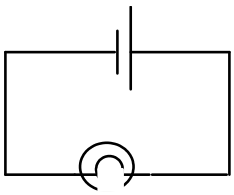
Explain why this happens to the other globe. _____

_____

c) Two globes in parallel will glow as brightly as just one globe i.e. the globes in circuit E will glow as brightly as the globe in circuit F. For a given cell, two globes in parallel provide less resistance and a so a greater current flow than two globes in series.



Circuit E



Circuit F

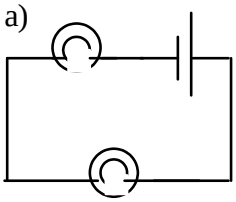
Explain why this happens _____

_____

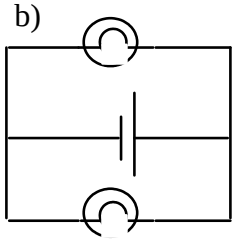
Example

Examine each of these circuit diagrams then answer the following questions.

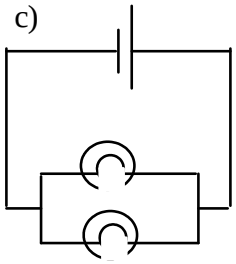
a)



b)



c)



1. In which of the following circuits are the globes in series, and in which are the globes in parallel?

_____

2. In which of the circuits will the globes glow the brightest and in which will they glow the dullest?

(Assume all cells and globes are identical)

_____

3. If in each circuit, one globe blows, will the second globe continue to glow? _____

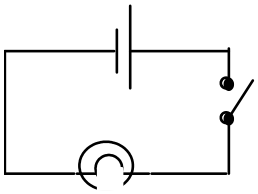
_____

ACTIVITY 6: Parallel Circuits

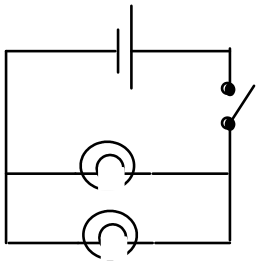
Method:

1. Construct ***Circuit E***, then C***ircuit F*** and compare the brightness of the globes in each.

E



F



Brightness of globes _____ _____

2. Describe the components of each of the circuits and the way they are connected in words

Circuit E contains _____

Circuit F contains _____

3. Construct C***ircuit F*** again . Press the switch on, then remove one of the globes from its socket. What happens to the other globe? _____

Explain your observation. _____

4. Construct C***ircuit B*** and C***ircuit F*** again, draw them below and then compare the brightness of the globes in each. Record your observations under each circuit diagram below..

Circuit B Circuit F

5. Complete the following:

For a given cell, 2 globes in parallel provide ..... resistance and glow ..... than 2 globes in series.

Challenging Connections:

6. Choose from one cell, a switch and three globes to construct a circuit in which the globe/s will be more dimly light than in any other possible circuit. Draw this circuit below.

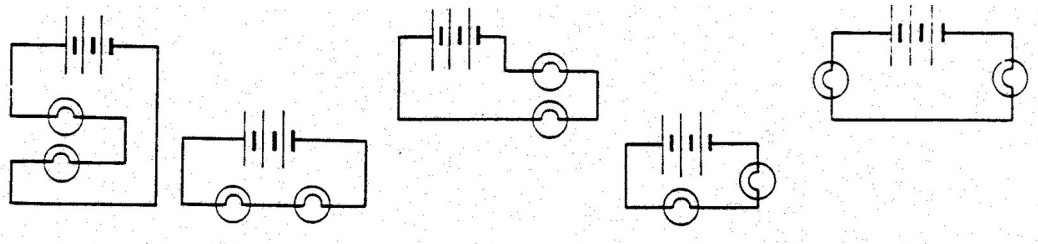
7. Choose from two cells, a switch and two globes to construct two circuits in which the globe/s will glow the brightest compared to any other possible circuit. Draw the two circuits below.

CIRCUIT 1 ***CIRCUIT 2***

8. From one cell, a switch and three globes construct a circuit in which one globe will glow brighter than the other two globes. Draw the circuit in the space below.

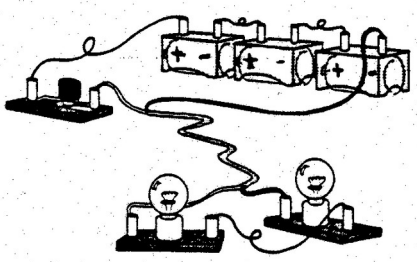
PROBLEM SHEET 3 - Series and Parallel Circuits.

1. Which of the following circuits show the globes connected in series?

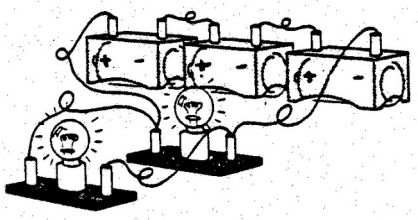


2. a) Draw the circuit diagram for each of the following circuits:

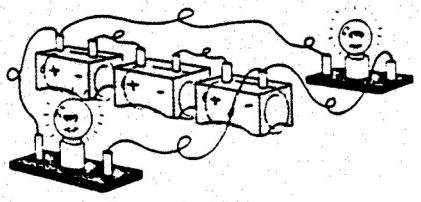
A



B



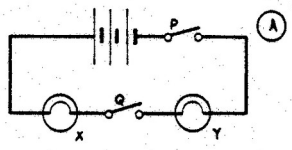
C



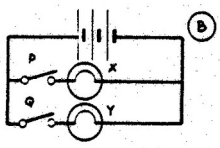
b) In which circuit will the globes glow dimmer than in the other circuits?

c) For each circuit, what will happen if one globe is removed?

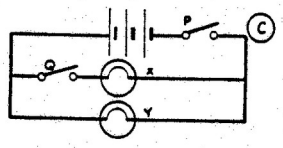
3. Complete the following tables for each circuit:



Switches		Lamps	
P	Q	X	Y
OFF	OFF		
OFF	ON		
ON	OFF		
ON	ON		

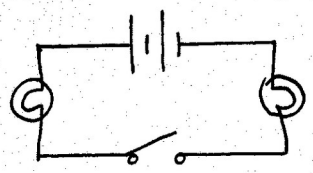


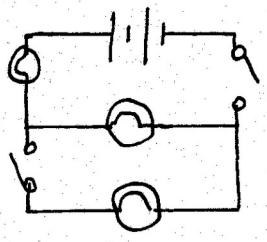
Switches		Lamps	
P	Q	X	Y
OFF	OFF		
OFF	ON		
ON	OFF		
ON	ON		

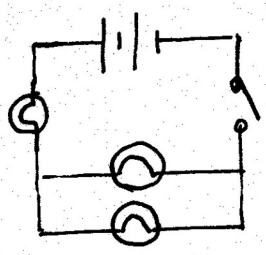


Switches		Lamps	
P	Q	X	Y
OFF	OFF		
OFF	ON		
ON	OFF		
ON	ON		

4. In which of these circuits are the lamps connected in parallel and in which are they connected in series?

A 

B 

C 

5. Draw the components for each of the circuit diagrams above.

6. Three globes are connected to a battery, but the connections cannot be seen.

   Globe A is brighter than B and C.

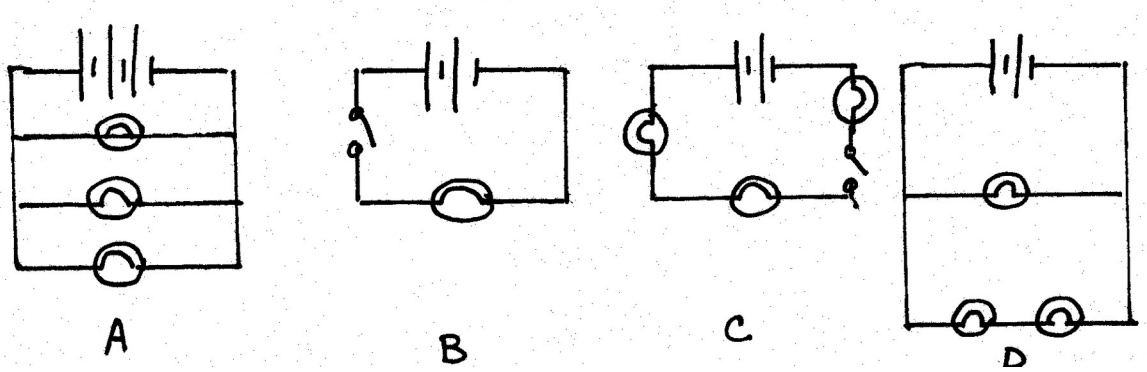
   When A is removed, B and C are not affected.

   When C is removed, B goes out but A shines on.

   a) Which two globes are connected in series?

   b) Which globe is in parallel with the other two?

   c) Draw the circuit

7. 

a) In which circuit would the globes be the brightest? Why?

b) In which circuit would the globes be the dimmest? Why?

c) Write down the circuits in order of globe brightness, brightest first.